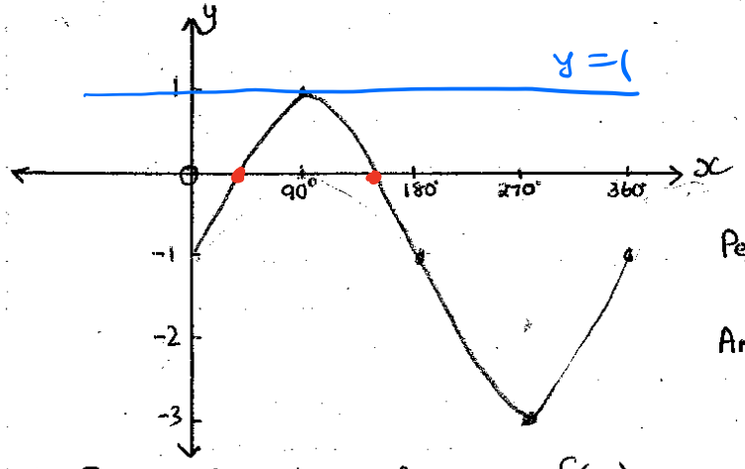
Grade 10 **Trig Graphs Worksheet**

① Sketch $f(x) = 2\sin x - 1$ $\quad x \in [0°; 360°]$



Range: $y \in [-3; 1]$

Period: 360°

Amplitude: 2

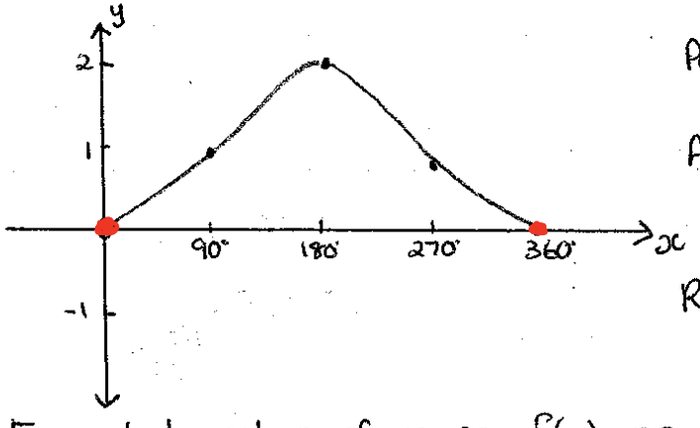
a) For which values of x is $f(x) = 0$?

30° & 150°

b) For which values of x is $f(x) = 1$?

90°

② Sketch $f(x) = -\cos x + 1$ $\quad x \in [0°; 360°]$



Period: 360°

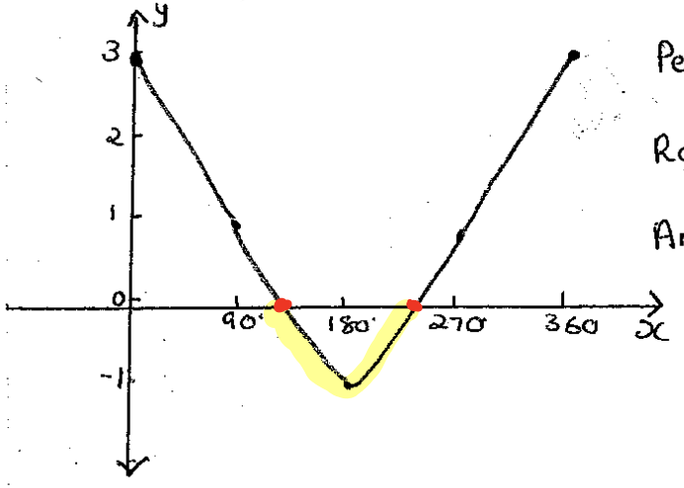
Amplitude: 1

Range: $y \in [0; 2]$

For which values of x is $f(x) = 0$?

0° & 360°

③ Sketch $f(x) = 2\cos x + 1$ $\quad x \in [0°; 360°]$



Period: 360

Range: $y \in [-1; 3]$

Amplitude: 2

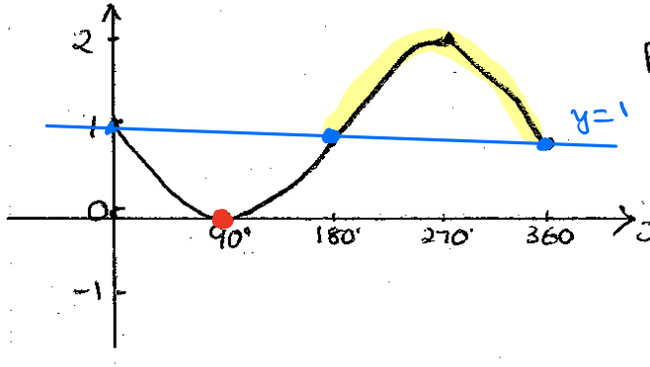
a) For which values of x is $f(x) = 0$?

135 & 225

b) For which values of x is $f(x) \leq 0$?

$x \in [135; 225]$

④ Sketch $f(x) = -\sin x + 1$ $\quad x \in [0°; 360°]$



Period: 360°

Range: $y \in [0; 2]$

Amplitude: 1

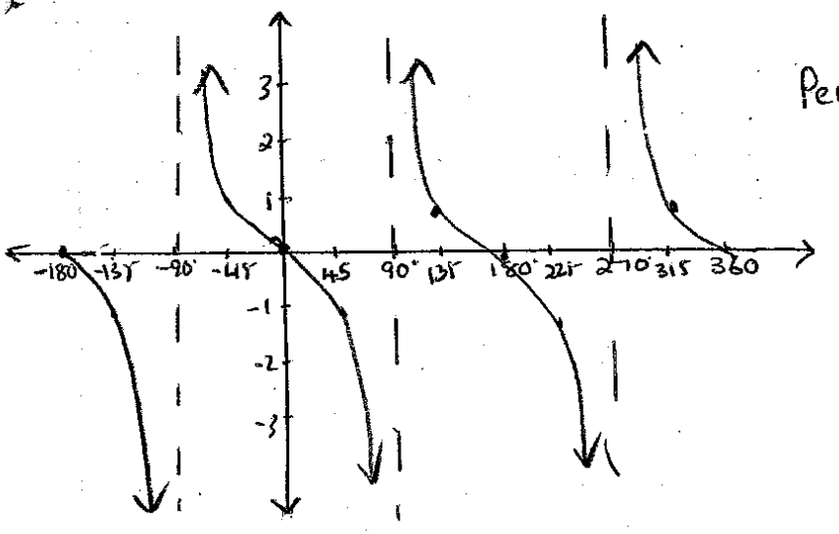
a) For which value(s) of x is $f(x) = 0$?

90°

b) For which values of x is $f(x) > 1$?

$x \in [180°; 360°)$

⑤ Sketch $f(x) = -\tan x$ $\quad x \in [-180°; 360°]$



Period: 180°

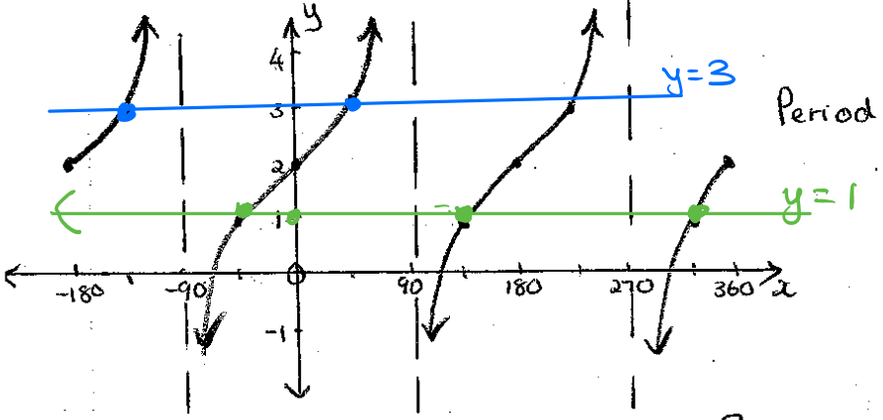
a) For which values of x is $f(x)$ undefined?

$-90°; 90°; 270°$

b) Give the equations of the asymptotes

$x = -90°$, $x = 90°$, $x = 270°$

⑥ Sketch $f(x) = \tan x + 2$ $\quad x \in [-180°; 360°]$



Period: 180°

a) For which values of x is $f(x) = 1$?

$-45°; 135°; 315°$

b) For which values of x is $f(x) = 3$? $-135°; 45°; 225°$

Trig Graphs: Translations and Reflections

⑦ Given: $f(x) = \sin x$ $\quad x \in [0°; 360°]$

Give the equation of h in each of the following:

a) $h(x)$ is the reflection of f in the x-axis

$h(x) = -\sin x$

b) $h(x)$ is the translation of f by $+2$ units vertical

$h(x) = \sin x + 2$

c) $h(x)$ is the stretch of f by a factor of 3.

$h(x) = 3\sin x$

d) $h(x)$ is the reflection of f in the x-axis and then translated $+1$ unit vertically.

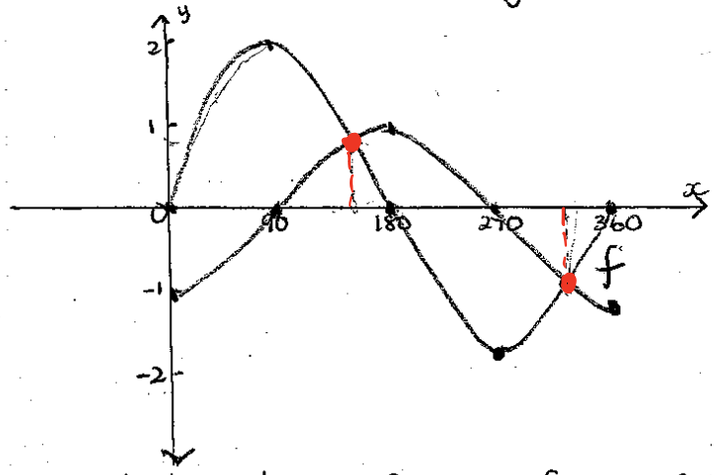
$h(x) = -\sin x + 1$

e) $h(x)$ is the stretch of f by a factor of 2 and then a shift upwards by 2 units

$h(x) = 2\sin x + 2$

⑧ Sketch on the same system of axes.

$f(x) = 2\sin x$ and $g(x) = -\cos x$ $\quad x \in [0°; 360°]$



For which values of x is $f(x) = g(x)$?